Recurrence Diagnostics for Stress Episodes: Clustering, Return Times, and Econometric Measurement

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July 2026

Abstract

Standard stress diagnostics in applied econometrics usually measure magnitude, tail loss, or event frequency, but not the temporal organization of extreme episodes. That omission matters whenever interpretation depends on whether extremes arrive as isolated shocks or as clustered stress. This paper develops a reproducible recurrence-oriented framework for stress diagnostics and introduces a cluster-adjusted stress functional that combines exceedance probability with a finite-threshold clustering coefficient. The contribution is deliberately bounded: a formal probabilistic setup, explicit estimators for exceedance incidence, return-time behavior, runs-based clustering, and multivariate stress activation, together with basic ordering and plug-in consistency results for the proposed functional under standard high-threshold regularity conditions. The asymptotic extremal-index literature enters as motivation and calibration, not as an object re-derived here. The synthetic evidence shows that recurrence diagnostics distinguish benchmark processes that volatility summaries alone describe only partially, though the reported rankings remain conditional on the stated thresholding and declustering choices. In the reported experiments, the estimated finite-level clustering coefficients for two stress benchmarks are 0.673 and 0.780, the logistic benchmark observable has normalized return-time coefficient of variation 0.795, the clustered synthetic benchmark has normalized return-time coefficient of variation 1.286, and rolling volatility is elevated during exceedance periods without exhausting the clustering information. The paper does not propose new asymptotic theory for extremal-index estimation and does not claim structural macroeconomic identification. Its contribution is a mathematically careful, computationally reproducible bridge between extreme-value diagnostics and econometric stress analysis.

Keywords: extreme values, extremal index, recurrence, return times, stress testing, time series econometrics

1 Introduction

The motivating problem is simple. In many macro-financial and econometric applications, a "stress episode" is treated as a period in which a variable becomes large in magnitude, volatile, or both. Yet stress is not described by magnitude alone. It also has a temporal shape. A series that produces isolated tail observations conveys a different kind of fragility from one that produces clustered extremes, even when the two have comparable event counts or similar rolling volatility.

The distinction is conceptually plain, but it is not often given explicit methodological form. Standard variance-based diagnostics, from ARCH and GARCH models onward, are designed to summarize time-varying second moments [8, 4]. Tail-risk and systemic-risk measures such as VaR, CoVaR, expected shortfall, and capital-shortfall metrics focus on the severity of bad states and their cross-sectional comovement [1, 2, 5, 3].

Yet the temporal spacing of threshold exceedances is a different object. Extreme-value theory for dependent sequences treats exceedance clustering and return times as first-order features of the data-generating process rather than as narrative afterthoughts [13, 7, 6]. The extremal index provides a canonical summary of tail clustering [15, 9, 16]. Related recurrence ideas also appear in the study of hitting times and rare events for stochastic and dynamical systems [11, 10, 14]. What is comparatively rare in applied econometrics is a workflow that makes these objects operational in a transparent, reproducible way and places them next to familiar volatility baselines rather than outside the analysis.

The gap addressed here is therefore narrower, and in a useful sense more exacting, than a generic shortage of methods. Extreme-value theory already provides a mature language for exceedance clustering, return times, and tail dependence, while econometric practice already offers sophisticated tools for volatility, downside risk, and financial stress. What remains underdeveloped is the middle ground between those domains: a framework in which recurrence itself is treated as part of what stress means and is analyzed alongside standard econometric summaries rather than left outside them. That is the problem taken up in this paper. I formalize a recurrence-based stress framework built from exceedance incidence, return-time behavior, runs-based clustering, multivariate stress activation, and volatility baselines; I implement those objects in a transparent computational pipeline that records intermediate artifacts rather than only final graphics; and I introduce a cluster-adjusted stress functional that distinguishes between stress processes with similar incidence but different temporal concentration of extremes. The theoretical contribution is narrow but real at the level claimed here: the functional is given a careful finite-threshold interpretation, an asymptotic motivation, basic ordering properties, and plug-in consistency under standard regularity conditions. The aim is not to compete with the asymptotic theory of dependent extremes on its own ground, but to make a disciplined part of that theory usable in applied stress diagnosis.

The paper does not claim more than the evidence can support. What is shown here comes from synthetic experiments and a small multivariate illustration, not from a structural macroeconomic application. The claim is correspondingly limited: recurrence diagnostics recover dimensions of stress persistence and organization that volatility summaries alone do not capture.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and identifies the specific gap addressed here. Section 3 introduces the probabilistic framework and the estimands. Section 4 describes the estimators and the implementation logic. Section 5 develops the cluster-adjusted stress functional and states basic theoretical results. Section 6 presents synthetic evidence from the generated artifacts. Section 7 discusses limitations and scope.

2 Literature Review

2.1 Extreme values under dependence

The first strand is the theory of extremes for dependent sequences. Classical EVT often begins with independent observations because independence isolates the basic tail mechanisms cleanly, but that simplification is rarely adequate for economic time series, where temporal dependence is often strongest in the turbulent periods that matter most. Leadbetter, Lindgren, and Rootzén [13] provided the foundational framework for extremes under dependence, while Embrechts, Klüppelberg, and Mikosch [7] and Coles [6] carried that framework into applied risk analysis. For the present paper, the lesson is straightforward: once dependence is admitted, the number of threshold exceedances no longer says everything that needs to be said about stress. Their temporal arrangement is part of the estimand.

2.2 Extremal index and clustering diagnostics

The second strand concerns the extremal index and the practical problem of quantifying clustering. O'Brien [15] established the extremal index as a scalar summary of local tail dependence for stationary sequences, and later work by Ferro and Segers [9] and Süveges [16] developed estimation strategies that made the object empirically usable while also showing how fragile inference can become under threshold choice, finite samples, and declustering rules. This literature is indispensable here, although I do not try to improve it on its own terms. The paper instead leans on it as established machinery and uses its outputs in a more design-conditional way than an estimation paper would: once threshold and declustering choices are fixed, the resulting clustering summaries are treated as diagnostics to be interpreted alongside other stress measures. The question is therefore more applied than theoretical: how should clustering diagnostics enter stress analysis so that they do not remain technical appendices to otherwise conventional volatility studies?

2.3 Recurrence and return-time analysis

The third strand studies recurrence itself, especially through hitting times and return-time distributions. In dynamical-systems and stochastic-process settings, these quantities have long been treated as structurally meaningful rather than merely descriptive. Hirata, Saussol, and Vaienti [11], together with Freitas, Freitas, and Todd [10] and Lucarini et al. [14], made that connection explicit for rare events and temporal dependence. The present paper does not try to transfer the strongest results of that literature directly into econometrics. What it borrows is the governing discipline: the spacing of rare events carries information distinct from their marginal severity and should therefore be measured rather than narrated after the fact. Applied econometrics has only partially absorbed that point.

2.4 Volatility econometrics and tail-risk baselines

The fourth strand is the econometric baseline against which the present framework should be understood. ARCH and GARCH models [8, 4] established time-varying volatility as a central empirical object, and they remain indispensable whenever the aim is to characterize changing second-moment conditions. Tail-risk measures and conditional downside summaries, including CoVaR [1], extend that baseline toward questions of severity and dependence in bad states. The point is not to dispute those tools, but to note that they answer a different question. They are designed to measure scale, exposure, and tail losses, whereas recurrence diagnostics are designed to measure the temporal organization of repeated exceedances.

2.5 Macro-financial stress monitoring and systemic-risk measurement

The fifth strand is the literature on stress monitoring and systemic risk. Surveys such as Bisias et al. [3] show how rich this literature already is in market-based, balance-sheet-based, and network-oriented indicators. Measures such as CoVaR [1], SRISK [5], and systemic expected shortfall [2] are designed to quantify the severity of distress, the associated capital shortfall, and the extent to which risk propagates across institutions. The present paper is not a competitor to that agenda. It targets a narrower diagnostic dimension that those measures do not place at the center, namely the recurrence profile of stress through time.

2.6 Gap and contribution

Taken together, these literatures point to a fairly precise opening. What is missing is not more general theory about dependent extremes, nor yet another volatility or systemic-risk measure. It is a framework in which recurrence, clustering, and return-time structure are treated as explicit stress objects and estimated within the same workflow as conventional econometric summaries. That is the contribution pursued in this paper. I formulate a recurrence-based stress framework in probabilistic terms, implement it through a deterministic computational pipeline, and use it to motivate a cluster-adjusted stress functional that preserves incidence information while penalizing temporal concentration of extremes. The mathematical contribution is intentionally limited: a finite-threshold functional, a small collection of basic properties, and an empirical workflow designed to keep interpretive claims tied to inspectable diagnostics. The result is not a new general theory of stress episodes, but a cleaner bridge between established theory on dependent extremes and the practical language of econometric stress analysis.

3 Probabilistic Framework

Let $\{X_t\}_{t\in\mathbb{Z}}$ be a strictly stationary real-valued process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with marginal distribution function F. The object X_t may represent a transformed return, a stress indicator, or another scalar measure whose upper tail corresponds to more severe conditions. The multivariate extension is introduced below.

For a threshold u, define the exceedance indicator

$$I_t(u) = \mathbf{1}\{X_t > u\}.$$

The exceedance probability is

$$p(u) = \mathbb{P}(X_0 > u).$$

In finite samples of length n, high-threshold analysis for estimation should work with an intermediate threshold sequence u_n satisfying

$$k_n := n\,\overline{F}(u_n) \to \infty, \qquad \frac{k_n}{n} \to 0,$$

where $\overline{F}(u) = 1 - F(u)$. This keeps exceedances rare but observable while still allowing their count to diverge. By contrast, the bounded-count regime

$$n\,\overline{F}(u_n) \to \tau \in (0, \infty)$$

belongs to point-process limit theory for rare events and is not the estimation regime used for the consistency claim below.

If $t_1 < \cdots < t_{N_n}$ denote the exceedance times among $\{1, \ldots, n\}$, the return times are

$$\tau_{j,n} = t_{j+1} - t_j, \qquad j = 1, \ldots, N_n - 1.$$

These random intervals summarize recurrence to the stress region. Their empirical distribution captures spacing, dispersion, and asymmetry in a way that event counts cannot.

To describe clustering, fix a run parameter $r \geq 1$. A new cluster begins at time t if $X_t > u_n$ and the previous r observations do not exceed u_n. The corresponding finite-level cluster-start indicator is[1]

$$C_{t,n}^{(r)} = I_t(u_n) \prod_{j=1}^{r} \left(1 - I_{t-j}(u_n)\right).$$

The associated finite-level clustering coefficient is

$$\theta_n^{(r)} = \frac{\mathbb{E}[C_{0,n}^{(r)}]}{\mathbb{E}[I_0(u_n)]} = \mathbb{P}(X_{-1} \leq u_n, \ldots, X_{-r} \leq u_n \mid X_0 > u_n).$$

When a $D^{(k)}$-type runs approximation is appropriate in the sense of O'Brien and subsequent declustering theory [15, 9], $\theta_n^{(r)}$ is a finite-level analogue of the extremal index: lower values indicate stronger local clustering of exceedances.

For a d-variate process $\{X_{t,j}\}$ with component-specific thresholds $u_{j,n}$, define

$$I_{t,j}(u_{j,n}) = \mathbf{1}\{X_{t,j} > u_{j,n}\}, \qquad S_t = \frac{1}{d} \sum_{j=1}^{d} I_{t,j}(u_{j,n}).$$

[1] The implementation uses a preceding-block definition. O'Brien's characterization is often written with a following block instead. For ergodic stationary sequences the two frequencies agree up to boundary terms, so the distinction is asymptotically negligible at the level used here.

The scalar $S_t \in [0,1]$ is a multivariate stress-state index: it records the share of active stress coordinates at time t under the chosen component thresholds. It is a descriptive screening device, not a causal object, and it discards the richer extremal-dependence structure that would be described by objects such as stable tail dependence functions or tail-dependence coefficients.

4 Estimators and Implementation

4.1 Sample estimators

Given observations $X_1, \ldots, X_n$, the empirical exceedance rate at threshold u_n is

$$\widehat{p}_n(u_n) = \frac{1}{n}\sum_{t=1}^{n} I_t(u_n).$$

The runs-based estimator of the finite-level clustering coefficient is

$$\widehat{\theta}_n^{(r)} = \frac{\sum_{t=r+1}^{n} C_{t,n}^{(r)}}{\sum_{t=1}^{n} I_t(u_n)},$$

whenever the denominator is positive. This estimator is simple, transparent, and computationally appropriate for a reproducible pipeline. It is not always the most statistically efficient choice, but it is easy to audit and aligns with the article's emphasis on inspectable artifacts. Because the numerator starts at $t = r+1$ while the denominator counts exceedances over all t, exceedances in the first r positions can only bias the finite-sample estimate downward.

The empirical return-time distribution is

$$\widehat{G}_n(x) = \frac{1}{N_n - 1}\sum_{j=1}^{N_n - 1} \mathbf{1}\{\tau_{j,n} \le x\},$$

provided $N_n \ge 2$. Summary functionals such as the mean, median, upper quantiles, and maximum of $\{\tau_{j,n}\}$ are used throughout the evidence section.

For the multivariate setting, the empirical stress-state index is

$$\widehat{S}_t = \frac{1}{d}\sum_{j=1}^{d} I_{t,j}(u_{j,n}),$$

and joint stress is defined by the event $\{\widehat{S}_t \ge 2/d\}$ in the present illustration. This threshold can be varied; the key point is to make the rule explicit, and in the present manuscript it should be read as a screening convention rather than as a structural threshold.

4.2 Threshold and run-length choices

Threshold selection is a substantive modeling decision, not a clerical preprocessing step. A threshold set too low destroys the rare-event interpretation; a threshold set too high produces severe finite-sample instability. The analysis therefore treats threshold sensitivity as part of the diagnostic output rather than as a hidden implementation detail. In applied work, one should report how clustering summaries move across a threshold grid and how the chosen run length r affects cluster identification.

The run parameter r has an equally direct interpretation. Larger values merge nearby exceedances into common clusters and therefore tend to increase measured persistence. This does not make large r "better." It means that r encodes what the analyst wishes to count as persistence. A serious stress paper should disclose that choice and test its sensitivity.

4.3 Repository implementation

The figures and source tables reported in this paper are generated by a deterministic implementation that proceeds from raw or simulated series to threshold definition, recurrence summaries, baseline econometric diagnostics, and article-facing outputs. The conceptual pipeline contains eight recorded stages: raw series, transformations, stress-region definition, exceedance events, clustering diagnostics, return-time diagnostics, econometric baselines, and article evidence. This separation matters because recurrence statistics are sensitive to thresholding and clustering rules; saved intermediate tables are therefore part of the method. Code and replication materials are available from the author.

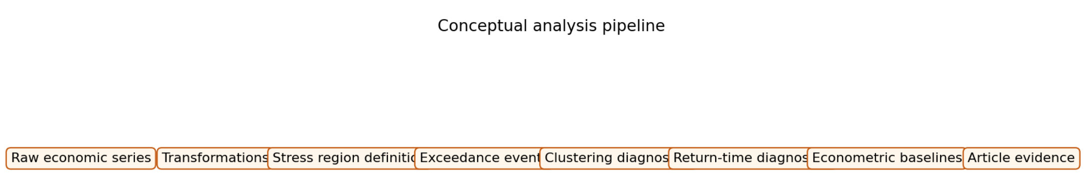

Figure 1: Conceptual pipeline for the computational workflow used in the paper. The methodological point is that recurrence objects are treated as explicit data products rather than informal visual impressions.

5 A Cluster-Adjusted Stress Functional

5.1 Definition

The theoretical contribution is deliberately narrow, but it is not merely notational. The aim is not to introduce a new asymptotic theory of extremes or a structural model of crisis propagation. It is to define a finite-threshold functional that records how tail incidence changes once local temporal concentration is taken seriously, and to do so in a form that survives contact with an auditable empirical pipeline. If recurrence is to count as part of stress measurement rather than as an auxiliary descriptive remark, then clustering information has to be translated into a quantity that can be compared across processes and thresholding schemes. At level n, both the marginal tail probability and the clustering coefficient are finite-threshold objects, so the functional introduced here is

$$\Lambda_n(u_n, r) = \frac{p(u_n)}{\theta_n^{(r)}}.$$

Its sample analogue is

$$\widehat{\Lambda}_n(u_n, r) = \frac{\widehat{p}_n(u_n)}{\widehat{\theta}_n^{(r)}},$$

whenever $\widehat{\theta}_n^{(r)} > 0$.

The interpretation becomes clear once the level of the object is kept in view. The finite-threshold runs coefficient $\theta_n^{(r)}$ should be read as a runs-based isolation coefficient, not as the literal reciprocal of mean cluster size. It measures how isolated exceedances are under the declustering rule indexed by r, and at finite thresholds it depends on both tail probability and run length. For that reason, $\Lambda_n(u_n, r)$ is not presented as an exact expected exceedance count or an exact probability. It is a stress functional that rescales exceedance incidence by lack of isolation, so for fixed $p(u_n)$ it rises when exceedances arrive in less isolated temporal configurations. The object is not a probability and can exceed one. The unemployment illustration below does exactly that, because the functional records cluster-weighted stress intensity rather than a frequency constrained to the unit interval. Because $\theta_n^{(r)} \leq 1$, the functional always satisfies $\Lambda_n(u_n, r) \geq p(u_n)$. At a finite threshold, however, the right no-clustering reference is not $p(u_n)$ itself but the independence benchmark

$$\Lambda_{\text{indep}}(u_n, r) = \frac{p(u_n)}{(1 - p(u_n))^r},$$

which exceeds $p(u_n)$ whenever $p(u_n) > 0$. That benchmark is not a secondary comparison; it is the finite-threshold no-clustering reference for Λ. In that sense, Λ may be read as a cluster-adjusted stress load: identical event frequencies need not imply identical stress severity if one process concentrates those events into more persistent bursts.

It is helpful to name the levels of analysis explicitly, because several nearby quantities appear in what follows. The coefficient $\theta_n^{(r)}$ is a finite-threshold runs-isolation coefficient defined at a chosen threshold and run length. The classical extremal index θ belongs to asymptotic rare-event theory and serves here as motivation rather than identification. The reported estimators $\widehat{\theta}_n^{(r)}$, $\widehat{\Lambda}_n(u_n, r)$, and their bootstrap intervals are fitted diagnostics attached to a particular sample and tuning configuration. Finally, $\Lambda_n(u_n, r)$ is the stress functional built from finite-threshold exceedance incidence and runs isolation. The asymptotic theory explains why clustering matters; the empirical sections below work at the finite-threshold diagnostic level unless stated otherwise.

The functional is also non-injective in a way that matters empirically. A rare but strongly clustered regime and a more frequent but weakly clustered regime can generate the same value of Λ. For that reason, Λ should never be reported in isolation. The pair $(\widehat{p}_n, \widehat{\theta}_n^{(r)})$ must accompany it whenever substantive interpretation is attempted.

The simplicity of the definition is intentional. The aim is not to compete with the richer theory of tail processes or extremal dependence structures, but to produce a scalar summary that remains mathematically interpretable and still usable inside an applied stress workflow.

5.2 Compound-Poisson motivation

The definition above also has an asymptotic motivation, but that motivation should not be mistaken for an exact finite-threshold identification. In the standard compound-Poisson limit for exceedance point processes under dependence; see Leadbetter, Lindgren, and Rootzén [13] for the general framework and Hsing, Hüsler, and Leadbetter [12] for the exceedance-point-process formulation. Fix a threshold sequence u_n such that

$$n\,p(u_n) \to \tau \in (0, \infty),$$

and let

$$N_n = \sum_{t=1}^{n} \mathbf{1}\{X_t > u_n\}\,\delta_{t/n}$$

denote the normalized exceedance point process. Under the usual mixing and anti-clustering conditions, N_n converges to a compound Poisson process

$$N = \sum_{j=1}^{\infty} \xi_j \delta_{\Gamma_j},$$

where the cluster centers Γ_j form a Poisson process on $[0, 1]$ with rate $\theta\tau$ and the marks ξ_j are i.i.d. cluster sizes with mean $\mathbb{E}\xi_j = 1/\theta$.

That asymptotic picture explains why a clustering adjustment is natural: in the rare-event limit, smaller extremal index means that a given amount of tail mass is being delivered through fewer and larger episodes. What it does *not* justify on its own is identifying the finite-threshold runs coefficient $\theta_n^{(r)}$ with the reciprocal of literal mean cluster size. The asymptotic extremal index from the point-process limit and the finite-threshold runs-isolation coefficient used in the empirical workflow are related only heuristically in the present paper. For that reason, $\Lambda_n(u_n, r)$ should be read as a finite-threshold incidence functional motivated by clustered exceedance limits, not as a theorem-level identity for expected exceedance counts.

5.3 Basic properties

Proposition 1 (Ordering under equal tail incidence)**.** *Fix a threshold u and run length r. Consider two stationary processes A and B such that*

$$p_A(u) = p_B(u) = p(u) > 0, \qquad \theta_A^{(r)}(u) < \theta_B^{(r)}(u) \le 1.$$

Then

$$\Lambda_A(u, r) > \Lambda_B(u, r).$$

Proof sketch. The claim follows immediately from the definition $\Lambda(u,r) = p(u)/\theta^{(r)}(u)$ and positivity of $p(u)$.

Proposition 2 (Finite-threshold independence benchmark)**.** *If exceedances are temporally independent at threshold u, then for every $r \geq 1$,*

$$\theta^{(r)}(u) = (1 - p(u))^r \qquad \text{and} \qquad \Lambda_{\text{indep}}(u,r) = \frac{p(u)}{(1-p(u))^r}.$$

If $\{u_n\}$ is a high-threshold sequence with $p(u_n) \to 0$, then

$$\theta_n^{(r)} = (1 - p(u_n))^r \to 1 \qquad \text{and} \qquad \frac{\Lambda_{\text{indep}}(u_n, r)}{p(u_n)} \to 1.$$

Proof sketch. Under independence,

$$\mathbb{P}(X_{-1} \leq u, \ldots, X_{-r} \leq u \mid X_0 > u) = \prod_{j=1}^{r} \mathbb{P}(X_{-j} \leq u) = (1 - p(u))^r,$$

so the relevant finite-threshold benchmark is exactly the probability that the preceding r observations remain below threshold under temporal independence. Substituting this value into the definition of Λ gives the finite-level independence benchmark, and the asymptotic statement follows because $p(u_n) \to 0$ implies $(1 - p(u_n))^r \to 1$.

Proposition 3 (Run-length monotonicity at fixed threshold)**.** *If u_n is fixed and $1 \leq r_1 < r_2$, then*

$$\theta_n^{(r_2)} \leq \theta_n^{(r_1)} \qquad \text{and} \qquad \Lambda_n(u_n, r_2) \geq \Lambda_n(u_n, r_1).$$

Proof sketch. If $r_2 > r_1$, then the event

$$\{X_{-1} \leq u_n, \ldots, X_{-r_2} \leq u_n\}$$

is a subset of

$$\{X_{-1} \leq u_n, \ldots, X_{-r_1} \leq u_n\},$$

so the monotonicity statement follows from conditional-probability ordering and the fixed numerator $p(u_n)$.

These propositions are modest in intent. They collect the algebraic consequences of the definition and identify the exact finite-threshold independence benchmark, but they do not claim a deeper structural theorem about clustered extremes.

Lemma 1 (Relative plug-in consistency under imported ingredients)**.** *Assume:*

- *(A1) $\{X_t\}$ is strictly stationary and satisfies dependence conditions under which the runs estimator is consistent at intermediate thresholds;*
- *(A2) u_n is an intermediate threshold sequence with $k_n = n\overline{F}(u_n) \to \infty$ and $k_n/n \to 0$;*
- *(A3) for a fixed run length r, the finite-level clustering coefficient $\theta_n^{(r)}$ is positive and bounded away from zero eventually;*

(A4) $\widehat{p}_n(u_n)/p(u_n) \xrightarrow{p} 1$ *and* $\widehat{\theta}_n^{(r)}/\theta_n^{(r)} \xrightarrow{p} 1$.

Then

$$\frac{\widehat{\Lambda}_n(u_n, r)}{\Lambda_n(u_n, r)} \xrightarrow{p} 1, \qquad \textit{equivalently} \qquad \widehat{\Lambda}_n(u_n, r)\frac{\theta_n^{(r)}}{p(u_n)} \xrightarrow{p} 1.$$

Proof sketch. Write

$$\frac{\widehat{\Lambda}_n(u_n, r)}{\Lambda_n(u_n, r)} = \frac{\widehat{p}_n(u_n)}{p(u_n)} \cdot \frac{\theta_n^{(r)}}{\widehat{\theta}_n^{(r)}}.$$

The first factor converges to one by assumption. The second does as well because $\widehat{\theta}_n^{(r)}/\theta_n^{(r)} \xrightarrow{p} 1$ and the reciprocal map is continuous on $(0, \infty)$. The continuous mapping theorem therefore yields the relative consistency claim. The purpose of the lemma is bookkeeping rather than originality: it does not prove consistency of the runs estimator. That ingredient is imported from the extremal-index literature under mixing and intermediate-threshold conditions; see, for example, [9, 16].

Taken together, the results in this subsection make a limited theoretical point. The four elementary propositions record ordering, benchmark, benchmark-normalization, and monotonicity properties at the finite-threshold level, and the plug-in lemma transfers consistency from established ingredients to the proposed functional. Their role is to keep the object mathematically legible, not to suggest that the paper resolves the harder theory of extremal-index estimation.

Proposition 4 (Finite-sample perturbation bound)**.** *Let* $p > 0$ *and* $\theta > 0$*, and let* $\bar{p}$ *and* $\bar{\theta}$ *be estimators such that*

$$|\bar{p} - p| \le \varepsilon_p, \qquad |\bar{\theta} - \theta| \le \varepsilon_\theta < \theta.$$

Then

$$\left|\frac{\bar{p}}{\bar{\theta}} - \frac{p}{\theta}\right| \le \frac{\varepsilon_p}{\theta - \varepsilon_\theta} + \frac{p\,\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)}.$$

Proof sketch. Write

$$\frac{\bar{p}}{\bar{\theta}} - \frac{p}{\theta} = \frac{\theta(\bar{p} - p) + p(\theta - \bar{\theta})}{\theta\bar{\theta}}.$$

Taking absolute values and using $\bar{\theta} \ge \theta - \varepsilon_\theta > 0$ gives

$$\left|\frac{\bar{p}}{\bar{\theta}} - \frac{p}{\theta}\right| \le \frac{\theta\varepsilon_p + p\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)},$$

which is the stated bound after separating terms. The interpretation is important: estimation error in the clustering term is amplified when θ is small, so cluster-adjusted stress is most numerically delicate precisely in the regimes where clustering is strongest.

This proposition is a deterministic sensitivity calculation, not a confidence statement or asymptotic approximation. Its value is to show where numerical fragility enters once a ratio functional is built from estimated tail incidence and estimated clustering.

Remark 1. *The functional* $\Lambda(u_n, r)$ *is best understood as a ranking and summary device, not as a welfare object or a structural sufficient statistic. Its value lies in comparative stress characterization.*

Remark 2. *The perturbation bound explains why threshold sensitivity cannot be treated as a cosmetic appendix item. If threshold changes move $\widehat{\theta}_n^{(r)}$ toward zero, the same sampling error in the clustering estimator induces a larger error in $\widehat{\Lambda}_n$. This is a concrete mathematical reason to report threshold and run-length robustness.*

6 Synthetic Evidence

6.1 Core numerical outputs

Table 1 collects the generated quantities used in the analysis.

Table 1: Core artifact summaries

Object	Numerical summary
Finite-level clustering coefficient by series (artifact 04; runs / intervals estimates for clustered synthetic, isolated synthetic, logistic benchmark observable)	Clustered synthetic stress: 0.673/0.721; isolated synthetic stress: 0.780/0.988; logistic benchmark observable: 1.000/1.000
Return-time diagnostics (artifact 03; logistic calibration in source table, clustered synthetic benchmark, Baa spread)	Logistic calibration: $\widehat{\theta}_{\text{runs}} = 1.000$, $\widehat{\theta}_{\text{int}} = 1.000$, CV 0.795; clustered stress: 0.673/0.721, CV 1.286; Baa spread: 0.186/0.213, CV 2.417
Multivariate stress timeline (artifact 05; coupled synthetic panel)	120 dated observations; 12 joint exceedance dates; maximum of 2 active series; mean stress score 0.150
Volatility comparison (artifact 06; macro-stress baseline series)	Mean rolling volatility during exceedances 0.3429; outside exceedances 0.2671; exceedance count 90

These quantities do not appear here as decorative summaries appended to a computational pipeline. They define the evidentiary boundary of the paper. Every substantive claim that follows is tied either to these numerical outputs or to the numerical tables underlying the corresponding figures. The resulting comparisons are therefore conditional on explicit threshold and run-length choices; they should be read as recurrence diagnostics under a stated design, not as structural rankings that would survive every admissible tuning decision. That restriction is methodologically important, because the manuscript is intended to argue from inspectable evidence rather than from narrative inflation.

6.2 Synthetic clustering benchmarks

The clearest synthetic comparison in the paper is the finite-level clustering-coefficient contrast between the two benchmark stress processes. At the reported threshold and run-length specification, the clustered benchmark yields an estimated coefficient of 0.673, whereas the more isolated benchmark yields 0.780. The gap is not large enough to support inflated language, but it is large enough to matter. What it captures is a difference in temporal organization that would be awkward to express, and easy to miss, if the discussion stayed at the level of generic instability or turbulence.

This is also the context in which the cluster-adjusted functional acquires substantive meaning rather than remaining a formal convenience. For comparable exceedance incidence under a fixed design, the series with the smaller clustering coefficient receives the larger stress load, because the same overall amount of tail activity is being delivered in a more temporally concentrated way. The theoretical section is therefore not separate from the synthetic evidence; it explains why this design-conditional ranking is worth measuring in the first place.

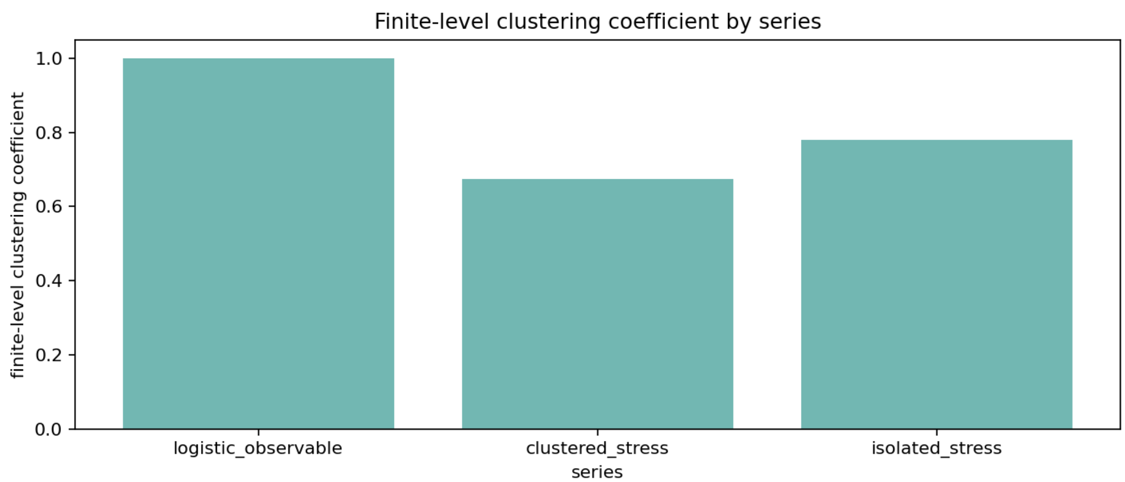

Figure 2: Estimated finite-level clustering coefficients across benchmark series. Lower values indicate stronger local clustering of exceedances. The result is finite-sample and threshold-dependent, but it recovers the intended ranking between clustered and more isolated stress processes.

6.3 Threshold robustness and uncertainty

Any recurrence-based stress diagnostic carries two kinds of uncertainty. One is ordinary sampling uncertainty: finite samples produce noisy exceedance counts, noisy cluster counts, and therefore noisy clustering summaries. The other comes from design choice: the reported statistic depends on the threshold and, for runs-based procedures, on the declustering rule. Robustness over a threshold grid is therefore not a cosmetic appendix exercise. It is part of the measurement problem itself. The analysis below treats it that way by pairing the threshold scan with a moving-block bootstrap band, so the path is read with an uncertainty envelope rather than as a string of isolated point estimates.

The level of interpretation should remain just as explicit here. The paths and intervals in this section describe the sampling behavior of fitted finite-threshold diagnostics under stated design choices; they are not direct measurements of an asymptotic extremal index. That distinction matters whenever a stable-looking path is turned into substantive language, because the bootstrap can quantify variation in the diagnostic while leaving structural interpretation conditional on the chosen thresholding and declustering scheme.

The threshold-sensitivity figure makes the point directly. The threshold scan uses run length $r = 4$ on the logistic benchmark observable

$$Y_t = -\log(|X_t - 1/2| + 10^{-12}), \qquad X_{t+1} = 4X_t(1 - X_t).$$

The distinguished point 1/2 is not periodic for the fully chaotic logistic map, so the periodic-point mechanism that can force extremal indices below one in dynamical-systems EVT is absent here [10]. The benchmark is therefore supposed to behave like a near-null calibration object. In that scan, the runs-based clustering estimate is 0.888 at the 0.90 quantile with 500 exceedances and rises to 0.909 at the 0.93 quantile with 350 exceedances. By the 0.95 quantile, the estimate reaches 1.000 with 250 exceedances and remains at that level through the 0.97 and 0.99 quantiles, even as the number of exceedances falls to 150 and then 50. Once the finite-level independence baseline is added, the correct reading becomes more precise. The corresponding i.i.d. benchmark values are $(1 - \widehat{p})^4 = 0.656$, 0.748, 0.815, 0.885, and 0.961, so the entire path lies above independence rather than below it. In other words, this scan is best read as a null-calibration exercise on a benchmark observable whose upper tail is already close to the finite-level no-clustering reference, not as direct evidence of persistent tail bunching.

That benchmark comparison is cleaner if the clustering estimate is renormalized directly. Define

$$\widetilde{\theta}_n^{(r)} = \frac{\widehat{\theta}_n^{(r)}}{(1 - \widehat{p}_n)^r}.$$

Under finite-threshold independence, the runs benchmark is exactly $(1 - \widehat{p}_n)^r$, so $\widetilde{\theta}_n^{(r)} = 1$ is the correct no-clustering reference at every threshold and run length. This is why $\widetilde{\theta}$ is a more honest cross-design comparison object than $\widehat{\theta}$ alone. In the logistic threshold scan, the renormalized values are 1.353, 1.215, 1.228, 1.130, and 1.041 across the 0.90, 0.93, 0.95, 0.97, and 0.99 quantiles. The path therefore confirms the same interpretation from a different angle: the benchmark observable is not showing excess clustering relative to independence. If anything, its finite-threshold exceedances are slightly more isolated than the exact i.i.d. runs benchmark, and that gap narrows as the threshold tightens.

That benchmarked statistic should still be read at the finite-threshold diagnostic level. It is not a structural clustering parameter in its own right. Its role is narrower: it removes the exact i.i.d. runs benchmark so that comparisons across thresholds and run lengths are not mechanically driven by the $(1 - \widehat{p}_n)^r$ term alone.

This threshold path also clarifies what uncertainty quantification should mean in a completed empirical application. A single point estimate of $\widehat{\theta}_n^{(r)}$ or $\widehat{\Lambda}_n(u_n, r)$ is not enough, because inferential uncertainty and threshold sensitivity interact. The bootstrap-enhanced figures and source tables move the analysis in the right direction: they do not solve the inferential problem, but they do show whether apparent stability at a given threshold reflects robust clustering evidence or simply the collapse of effective sample size. The same logic now extends to the cluster-adjusted stress functional itself. In the current threshold scan, the estimated functional falls from 0.113 at the 0.90 quantile to 0.077 at 0.93, then to 0.050, 0.030, and 0.010 at the 0.95, 0.97, and 0.99 quantiles, respectively, while the corresponding finite-level independence benchmark falls from 0.152 to 0.094, 0.061, 0.034, and 0.010. The associated 90% block-bootstrap intervals are $[0.104, 0.118]$, $[0.069, 0.083]$, $[0.044, 0.054]$, $[0.025, 0.034]$, and $[0.008, 0.013]$. The important point is not merely that $\widehat{\Lambda}$ declines. It is that the decline largely tracks the finite-threshold mechanical benchmark, which is exactly what one should expect from a null-calibration observable with little excess clustering beyond the $(1-\widehat{p})^r$ effect. Read that way, the threshold path is a sensitivity diagnostic for a family of finite-threshold objects, not an estimate of a single structural quantity unfolding across quantiles.

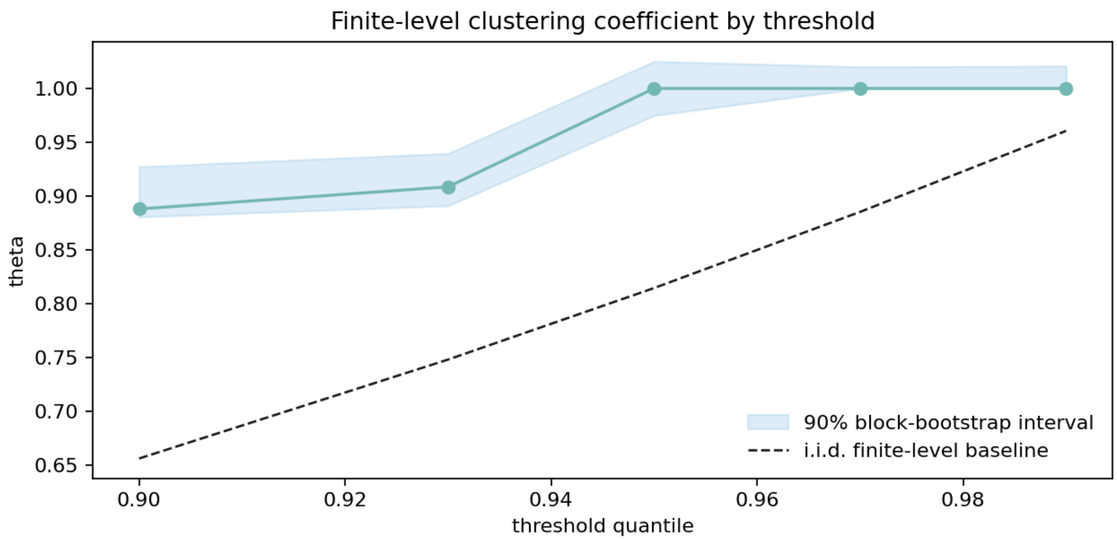

Figure 3: Threshold sensitivity of the clustering estimate. The dashed curve is the finite-level i.i.d. benchmark $(1-\widehat{p})^r$, so the figure shows not only how the estimate moves with the threshold, but also whether it departs materially from the no-clustering reference at each cutoff.

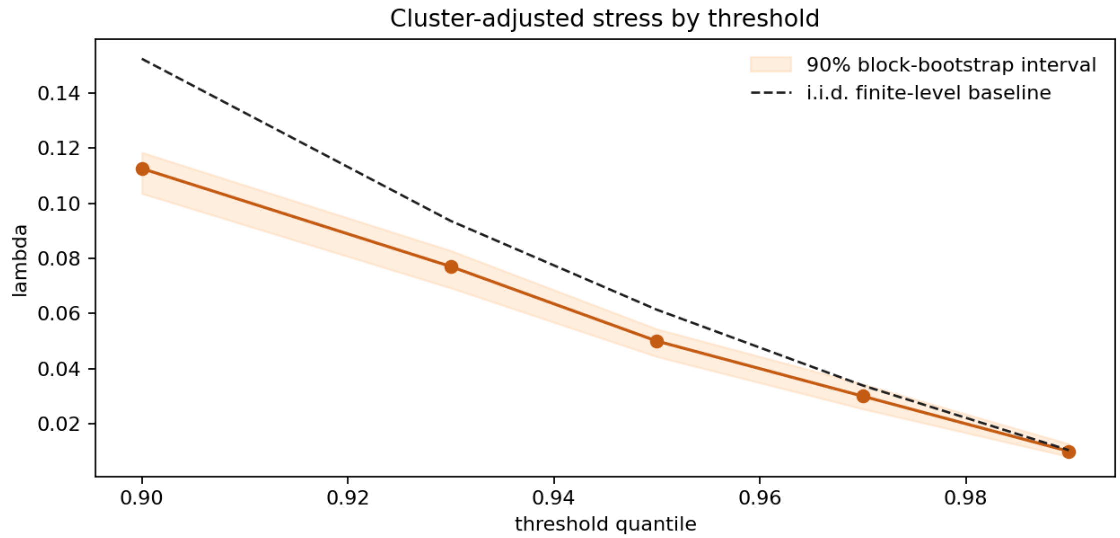

Figure 4: Threshold sensitivity of the cluster-adjusted stress functional. The dashed curve is the finite-level independence benchmark $\widehat{p}/(1-\widehat{p})^r$, which is the honest reference for deciding whether the observed path reflects excess clustering or merely the mechanical $(1-\widehat{p})^r$ adjustment.

Run-length sensitivity provides a second robustness check because the declustering rule is itself part of the stress definition. At the fixed 0.90 threshold quantile used in the run-length scan on the same logistic benchmark observable, the estimated cluster-adjusted functional equals 0.100 for run lengths 1 and 2, then rises to 0.113, 0.149, and 0.206 for run lengths 4, 6, and 8, respectively. The corresponding 90% block-bootstrap intervals are $[0.095, 0.104]$, $[0.094, 0.104]$, $[0.104, 0.117]$, $[0.134, 0.159]$, and $[0.178, 0.225]$. Taken on their own, those numbers might invite a persistence interpretation, but that reading is false. The finite-level independence benchmark at $\widehat{p} = 0.10$ is $\widehat{\Lambda}_{\text{indep}} = 0.111$, 0.123, 0.152, 0.188, and 0.233 for $r = 1$, 2, 4, 6, and 8, respectively, so the observed path sits at or below independence throughout. The increase in $\widehat{\Lambda}$ with r is therefore mechanical rather than diagnostic: it occurs under i.i.d. data as well because $(1-\widehat{p})^r$ falls with r. Different run lengths here define different finite-threshold diagnostics by construction, so the path should be read as a sensitivity profile over admissible declustering choices rather than as movement in a single latent persistence parameter. The substantive quantity is the gap to the independence reference, not raw monotonicity in the run length.

The renormalized clustering path makes the same point with less room for confusion. At the same threshold, $\widetilde{\theta}$ equals 1.111, 1.235, 1.353, 1.261, and 1.129 for $r = 1$, 2, 4, 6, and 8. Those values are not constant, so renormalization does not magically remove all finite-sample variation, but it does remove the false intuition that a rising raw $\widehat{\Lambda}$ or a falling raw $(1-\widehat{p})^r$ is automatically evidence of more persistence. The benchmark observable stays above one throughout, which is exactly what the earlier threshold scan already suggested: this is a calibration object whose exceedances are not more tightly bunched than the finite-level independence reference.

Figure 5: Run-length sensitivity of the cluster-adjusted stress functional at the 0.90 threshold quantile. The dashed curve is the finite-level i.i.d. benchmark, so the figure makes clear that raw monotonicity in r carries no persistence information by itself; only departures from the benchmark do.

Table 2: Compact robustness summary for the cluster-adjusted stress functional

Path	Setting	$\widehat{\Lambda}$	$\widehat{\Lambda}_{\text{indep}}$	90% bootstrap interval
Threshold	$q = 0.90$	0.113	0.152	[0.104, 0.118]
Threshold	$q = 0.93$	0.077	0.094	[0.069, 0.083]
Threshold	$q = 0.95$	0.050	0.061	[0.044, 0.054]
Threshold	$q = 0.97$	0.030	0.034	[0.025, 0.034]
Threshold	$q = 0.99$	0.010	0.010	[0.008, 0.013]
Run length	$r = 1$	0.100	0.111	[0.095, 0.104]
Run length	$r = 2$	0.100	0.123	[0.094, 0.104]
Run length	$r = 4$	0.113	0.152	[0.104, 0.117]
Run length	$r = 6$	0.149	0.188	[0.134, 0.159]
Run length	$r = 8$	0.206	0.233	[0.178, 0.225]

6.4 Return-time diagnostics through the Ferro–Segers mixture

The return-time discussion becomes more coherent once it is organized around the same clustering object used elsewhere in the paper. Under local dependence and the usual rare-event scaling, the normalized inter-exceedance times do not simply target a unit exponential law. Their limiting benchmark is the Ferro–Segers mixture

$$(1 - \theta)\delta_0 + \theta\,\mathrm{Exp}(\theta),$$

whose atom at zero records, at the limiting level, the mass generated by within-cluster recurrences and whose exponential tail describes spacing between clusters. That asymptotic statement should be kept separate from what is done empirically here. In this paper, the mixture is used only as a fitted diagnostic reference built from the estimated clustering coefficient; it is not asserted as an exact finite-sample law for observed waiting times. The evidence for clustering therefore comes from the combination of estimated coefficients, dispersion summaries, and QQ geometry relative to that fitted reference, not from a claim that any one sample literally follows the limiting mixture.

The logistic benchmark observable remains useful precisely because it collapses back to the null-like case. At threshold quantile 0.95 with run length $r = 4$, both the runs estimator and the intervals estimator equal 1.000. The artifact contains 249 inter-exceedance intervals, the normalized coefficient of variation is 0.795, the KS distance from the unit exponential benchmark is 0.221, and the moving-block bootstrap reference value is 0.056. Read properly, that panel is not supposed to be dramatic. It is a calibration object showing what the fitted return-time diagnostic looks like when the zero-mass component is negligible and the recurrence pattern stays close to the no-clustering reference. The KS number is therefore descriptive rather than a formal goodness-of-fit test for an exact exponential model.

The clustered synthetic benchmark is the more revealing case. Using the absolute GARCH$(1, 1)$ benchmark at the same threshold quantile and run length gives $\widehat{\theta}_{\text{runs}} = 0.673$ and $\widehat{\theta}_{\text{int}} = 0.721$, together with 149 return times, a median waiting time of 10.0, a 90th percentile of 53.6, a maximum of 145, and a normalized coefficient of variation of 1.286. The gap between the two estimators is small enough to be reassuring rather than troublesome: both identify substantial local clustering. Once the QQ panel is read against the fitted mixture instead of against a pure exponential line, its geometry becomes more interpretable. Excess short waits are consistent with repeated exceedances arriving inside the same burst, while the stretched right tail is consistent with quieter gaps separating one burst from the next. That is a fitted-diagnostic reading of the sample, not a proof that the finite series has already entered its limiting regime.

The empirical Baa–Treasury spread sits between the logistic calibration and the deliberately clustered synthetic process, which is why it makes a useful second panel. At threshold quantile 0.90 with run length $r = 3$, the series yields $\widehat{\theta}_{\text{runs}} = 0.186$ and $\widehat{\theta}_{\text{int}} = 0.213$, so the fitted mixture assigns atom mass $1 - \widehat{\theta}_{\text{int}} = 0.787$ to near-immediate recurrence in the limiting reference picture. Its normalized coefficient of variation is 2.417, much larger than in the logistic calibration and larger even than in the synthetic clustered benchmark. I do not want to over-interpret a single monthly spread series, and this diagnostic does not identify a structural mechanism for credit stress. The point of the panel is narrower: the same fitted-mixture language remains readable outside simulation, and it captures an empirical recurrence pattern with short bunches of exceedances separated by longer quieter stretches.

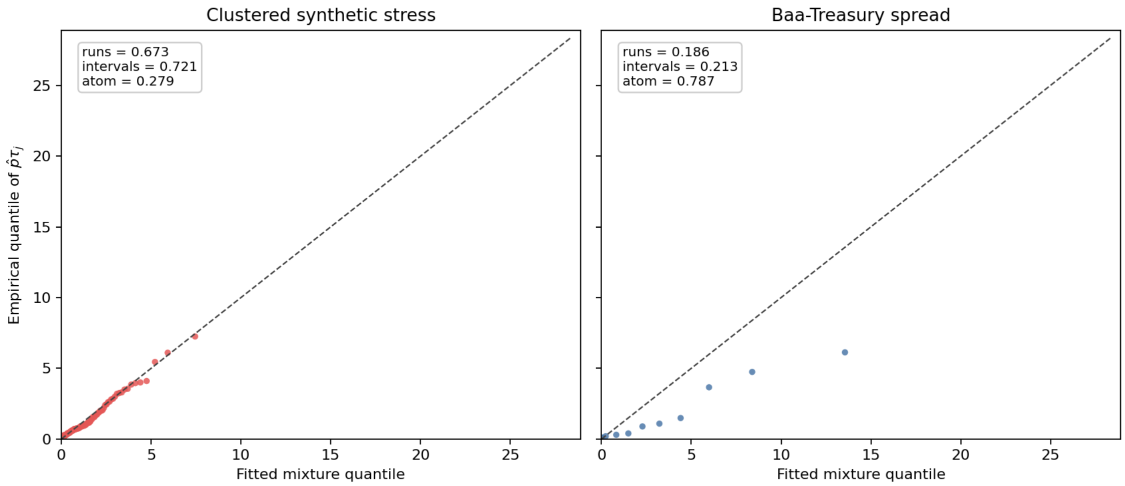

Figure 6: QQ comparison of normalized return times $\widehat{p}\tau_j$ against the fitted Ferro–Segers mixture. The clustered synthetic benchmark shows the expected excess of short recurrence gaps and a stretched right tail, while the Baa–Treasury spread exhibits the same geometry in milder but still visible form.

6.5 Multivariate stress activation

The multivariate illustration records 120 observations and 12 joint exceedance dates, with at most two simultaneously active series and a mean stress score of 0.150. Those counts are specific to the present panel and threshold specification, but the point they support is simple and worth making explicitly. Stress is not distributed uniformly across coordinates through time: some episodes are effectively univariate, while others involve simultaneous activation across variables. That distinction is descriptive rather than structural, but it ought to be measured before stronger language about contagion, propagation, or systemic amplification enters the discussion.

The stress-state index is therefore best understood as a screening device for joint activation rather than as a substitute for structural multivariate analysis. Its purpose is to mark the temporal geometry of shared stress under a declared exceedance design, not to explain why that geometry arises or to provide a system-wide dependence parameter.

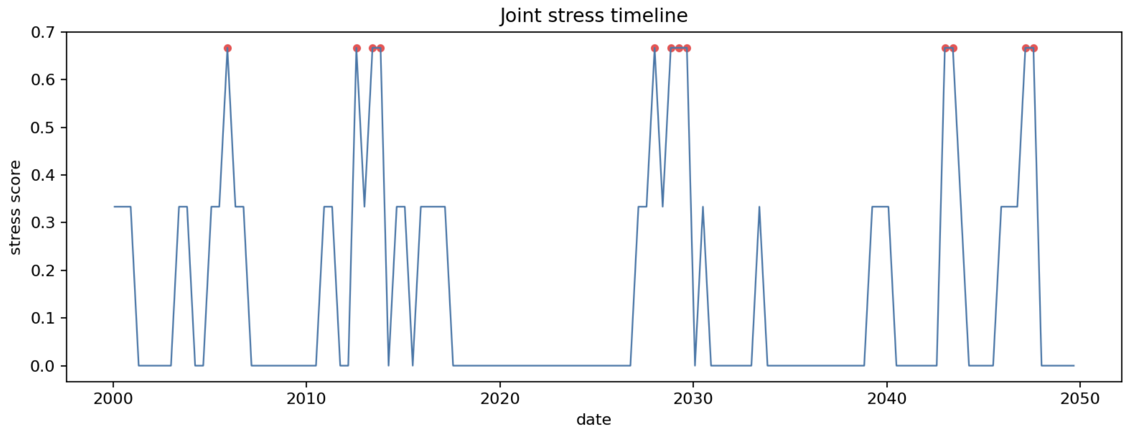

Figure 7: Multivariate stress-state timeline. The figure does not identify transmission channels; it records when stress is isolated and when it is joint across observables.

6.6 Why the volatility baseline remains necessary

The baseline comparison shows higher mean rolling volatility during exceedance periods (0.3429) than outside them (0.2671). This is both substantively useful and entirely unsurprising. A recurrence framework should not be judged by whether it overturns established volatility evidence, because that is not the standard it is meant to meet. Its purpose is to recover information that volatility summaries do not encode directly.

The conclusion is one of complementarity rather than rivalry. Volatility remains the natural object when the question concerns changing scale conditions, while recurrence diagnostics matter when the question concerns how extreme states reappear and bunch together through time. A stress workflow that excludes either object leaves something important out.

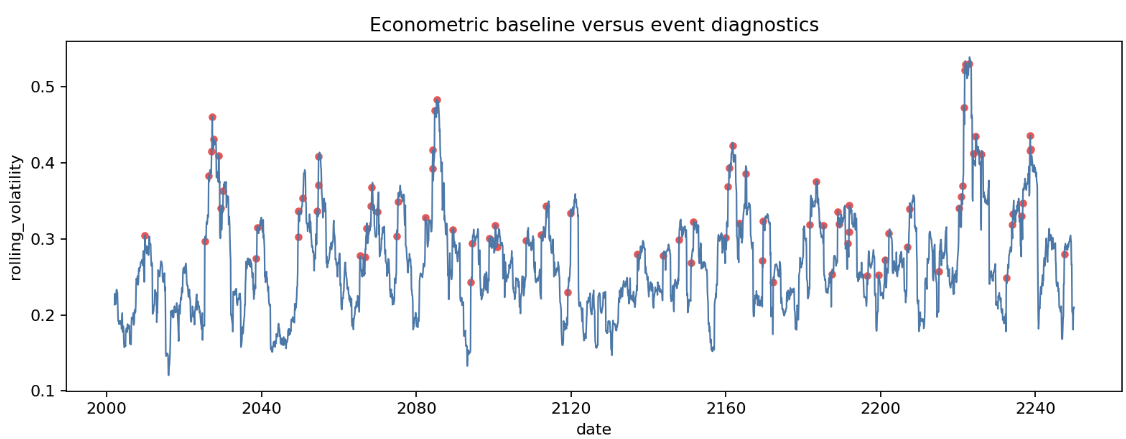

Figure 8: Baseline econometric and recurrence-oriented diagnostics are complementary. Elevated volatility during exceedance periods is informative, but it does not encode clustering intensity or return-time dispersion.

6.7 A small empirical macro-financial panel

The paper now includes a genuine, if still modest, empirical illustration built from three publicly available monthly FRED series rather than from a toy bundled example. The panel consists of the civilian unemployment rate, the Moody's Baa minus 10-year Treasury spread, and the Kansas City Financial Stress Index. After aligning the cleaned local CSV extracts on their common support, the usable sample runs from February 1990 through May 2026 and contains 435 monthly observations. For comparability with the synthetic diagnostics, the baseline empirical specification uses upper-tail exceedances at the 0.90 quantile together with a runs declustering parameter of $r = 3$. Because the unemployment level is visibly dominated by low-frequency regime shifts, I also report a 12-month difference for that series; the transformation removes slow-moving level changes while preserving a monthly cyclical interpretation.

At that baseline specification, the unemployment level row is best treated as descriptive only. Its threshold is 8.2; it produces 42 exceedances, but those exceedances collapse into only 2 runs clusters, yielding $\widehat{\theta}_{\text{runs}} = 0.048$, $\widehat{\theta}_{\text{int}} = 0.049$, and $\widehat{\Lambda} = 2.028$. Those numbers mainly restate a familiar historical fact: the sample contains two long recessionary unemployment episodes, and the stationarity and mixing assumptions from Section 3 are not credible enough on the level scale to support much inferential weight. The transformed unemployment series is more useful for comparison. At the same threshold quantile and run length, the 12-month difference has threshold 1.30, 42 exceedances, 4 runs clusters, $\widehat{\theta}_{\text{runs}} = 0.095$, $\widehat{\theta}_{\text{int}} = 0.142$, and $\widehat{\Lambda} = 1.043$. The two financial indicators remain less concentrated by point estimate: the Baa spread crosses its threshold of 3.16 in 43 months grouped into 8 clusters, giving $\widehat{\theta}_{\text{runs}} = 0.186$, $\widehat{\theta}_{\text{int}} = 0.213$, and $\widehat{\Lambda} = 0.531$, while KCFSI crosses its threshold of 1.068 in 44 months also grouped into 8 clusters, giving $\widehat{\theta}_{\text{runs}} = 0.182$, $\widehat{\theta}_{\text{int}} = 0.161$, and $\widehat{\Lambda} = 0.556$. The empirical point is more restrained than in the earlier draft. Level unemployment stress is concentrated in a few long blocks, but the comparison that can actually bear interpretation is the one between transformed unemployment and the financial indicators. The difference remains visible, though no longer in a form that justifies dramatic ranking claims. It should also be read as design-conditional: changing the threshold or the declustering rule changes the numerical distance, even when the broader recurrence pattern remains recognizable. The intervals estimator now appears beside the runs estimator because it gives a second view of the same recurrence geometry; here it slightly enlarges the transformed unemployment and Baa estimates, lowers KCFSI modestly, and leaves the broad ranking intact.

The bootstrap treatment has also been corrected. The earlier percentile intervals mixed threshold re-selection into what was supposed to be fixed-design sampling uncertainty, and for low-cluster cases they produced exactly the kind of boundary pathology that should have triggered rejection rather than reporting. The revised empirical intervals condition on the original numeric threshold, use basic bootstrap reflection, report a sensitivity envelope over block sizes $\{12, 24, 36, 48\}$ rather than a single ad hoc block choice, and declare rows with fewer than 3 clusters not interpretable instead of forcing a numeric band. Those envelopes are built from nominal 90% block-bootstrap intervals, but the envelope itself should not be read as carrying a literal 90% coverage guarantee once the outer band across block sizes is taken. That is why the unemployment level row in Table 3 now carries an explicit degeneracy note rather than a misleading confidence band. For the transformed unemployment series, the sensitivity envelopes built from 90% block-bootstrap intervals are $[0, 0.190]$ for $\widehat{\theta}_{\text{runs}}$ and $[0, 1.934]$ for $\widehat{\Lambda}$. For the Baa spread, they are $[0.002, 0.213]$ and $[0.093, 0.903]$; for KCFSI, $[0, 0.267]$ and $[0, 1.043]$. These are wide bands, and they should be read that way. The transformed unemployment point estimate remains larger than the two financial estimates, but interval overlap is substantial, so the uncertainty treatment supports caution rather than a hard ranking.

A small synthetic coverage harness was added to check that the repaired interval construction is at least numerically coherent on the clustered benchmark used elsewhere in the paper. Relative to a long-run benchmark reference with $\theta \approx 0.646$ and $\Lambda \approx 0.155$, the nominal 90% basic bootstrap intervals covered the benchmark in 0.875 of replications for θ and 1.000 for Λ across a run of 40 Monte Carlo samples. Those figures should not be read too aggressively: with only 40 replications they are illustrative rather than definitive, and they do not establish finite-sample validity. What they do show is narrower but still useful. The revised procedure is no longer failing the elementary self-checks that the earlier intervals did.

Table 3: Empirical clustering summary for the monthly FRED panel, with sensitivity envelopes built from block-bootstrap intervals

Series	$\widehat{\theta}_{\text{runs}}$	$\widehat{\theta}_{\text{int}}$	sensitivity envelope	$\widehat{\Lambda}$	sensitivity envelope
Unemployment rate (level)	0.048	0.049	not interpretable ($k = 2$)	2.028	not interpretable ($k = 2$)
Unemployment rate (12m diff.)	0.095	0.142	[0, 0.190]	1.043	[0, 1.934]
Baa-Treasury spread	0.186	0.213	[0.002, 0.213]	0.531	[0.093, 0.903]
KCFSI	0.182	0.161	[0, 0.267]	0.556	[0, 1.043]

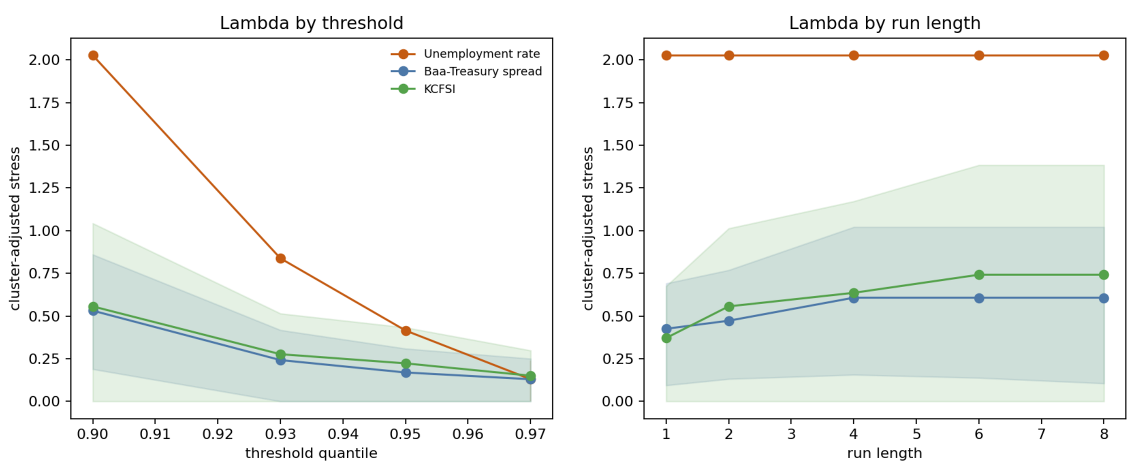

Figure 9: Empirical robustness of the cluster-adjusted stress load. The level-unemployment path should be read descriptively because it is generated by only two baseline clusters; the financial indicators remain the more interpretable comparison objects under threshold and run-length variation.

The robustness figure is still useful, but its interpretation must now respect the inferential split. Along the threshold dimension, level unemployment is the most sensitive series: its estimated stress load falls from 2.028 at $q = 0.90$ to 0.838, 0.415, and 0.130 at $q = 0.93$, 0.95, and 0.97, respectively. That compression is descriptive evidence that the broader upper-tail unemployment regime is dominated by a few long episodes, whereas the most extreme observations are less tightly packed. The Baa spread and KCFSI decline more smoothly, moving from 0.531 to 0.130 and from 0.556 to 0.150 over the same threshold grid. Along the run-length dimension, level unemployment is essentially invariant because its two major stress episodes are already separated by long calm intervals, while the two financial indicators become progressively more clustered as the episode definition widens. For the Baa spread, $\widehat{\Lambda}$ rises from 0.425 at $r = 1$ to 0.607 by $r = 4$; for KCFSI it rises from 0.371 at $r = 1$ to 0.742 by $r = 6$. The transformed unemployment specification is not plotted in that figure because its purpose is inferential repair rather than visual comparison, but its baseline point estimate sits between the level-unemployment and financial-series readings. The corrected conclusion is therefore conditional: labor-market stress is the most concentrated object descriptively in levels, yet the inferentially usable differences against the financial indicators are materially less sharp once the transformation and interval repair are imposed.

The renormalized clustering diagnostic sharpens that reading further. Because $\widetilde{\theta} = 1$ under independence at every finite threshold and run length, values below one indicate more bunching than the exact i.i.d. benchmark and values above one indicate more isolation. At the baseline empirical specification $(q, r) = (0.90, 3)$, the renormalized runs estimates are 0.065 for unemployment in levels, 0.130 for transformed unemployment, 0.254 for the Baa spread, and 0.250 for KCFSI. That ranking is more stable than the raw $\widehat{\Lambda}$ ranking because it removes the mechanical effect of changing $(1 - \widehat{p})^r$ across designs.

Even here the interpretation should remain design-conditional. The statistic is not a threshold-free measure of structural clustering; it is a finite-threshold comparison against the exact independence benchmark generated by the chosen exceedance design.

It also answers the run-length question more cleanly than the raw robustness plot. The earlier visual pattern suggested a reversal: level unemployment looked flat in r, whereas the financial indicators rose as the declustering window widened. That statement does not survive renormalization literally. Once the exact independence benchmark is divided out, the level-unemployment path is no longer flat at all; it rises from 0.053 to 0.058, 0.071, 0.088, and 0.107 as r moves from 1 to 2, 4, 6, and 8. The Baa spread takes values 0.258, 0.258, 0.247, 0.304, and 0.374, while KCFSI takes 0.303, 0.225, 0.244, 0.259, and 0.320. So the sharp raw reversal disappears. What remains is the more substantive statement: all three series stay below one throughout, and unemployment remains the most clustered object relative to the finite-threshold independence benchmark even after the mechanical run-length effect is removed.

Table 4: Compact empirical robustness summary for the cluster-adjusted stress load

Series	Threshold path $\widehat{\Lambda}(q, 3)$	Run-length path $\widehat{\Lambda}(0.90, r)$
Unemployment rate	$2.028 \to 0.838 \to 0.415 \to 0.130$	$2.028 \to 2.028 \to 2.028 \to 2.028 \to 2.028$
Baa-Treasury spread	$0.531 \to 0.242 \to 0.169 \to 0.130$	$0.425 \to 0.472 \to 0.607 \to 0.607 \to 0.607$
KCFSI	$0.556 \to 0.276 \to 0.223 \to 0.150$	$0.371 \to 0.556 \to 0.636 \to 0.742 \to 0.742$

Joint activation sharpens the historical interpretation. Defining a joint stress month as one in which at least two of the three indicators exceed their respective thresholds, the panel contains 29 such months and an average stress score of 0.099, with all three indicators active simultaneously at the peak. The dominant cluster runs from August 2008 through August 2009, which is precisely the kind of prolonged, multi-margin stress episode the recurrence framework is meant to distinguish from short-lived turbulence. Smaller joint clusters appear around the 2001 recession, the 2011 euro-area stress period, and the initial pandemic shock in March through May 2020. The empirical point is not that these dates are surprising; it is that the framework measures them in a way that keeps incidence, clustering, and joint activation analytically separate.

This remains a restrained empirical section rather than a full application. The thresholds are deliberately simple, no structural identification is attempted, and the evidence should not be read as a claim about optimal early-warning design. Even so, the illustration is now doing the right kind of work for a methods paper. It shows that recurrence diagnostics survive contact with an interpretable macro-financial panel, but also that they force discipline about when a dramatic point estimate is only descriptive and when an inferential comparison is actually defensible. The main empirical distinction remains the contrast between long labor-market stress blocks and more episodic, design-sensitive financial turbulence, but the uncertainty repair makes that contrast narrower and more credible.

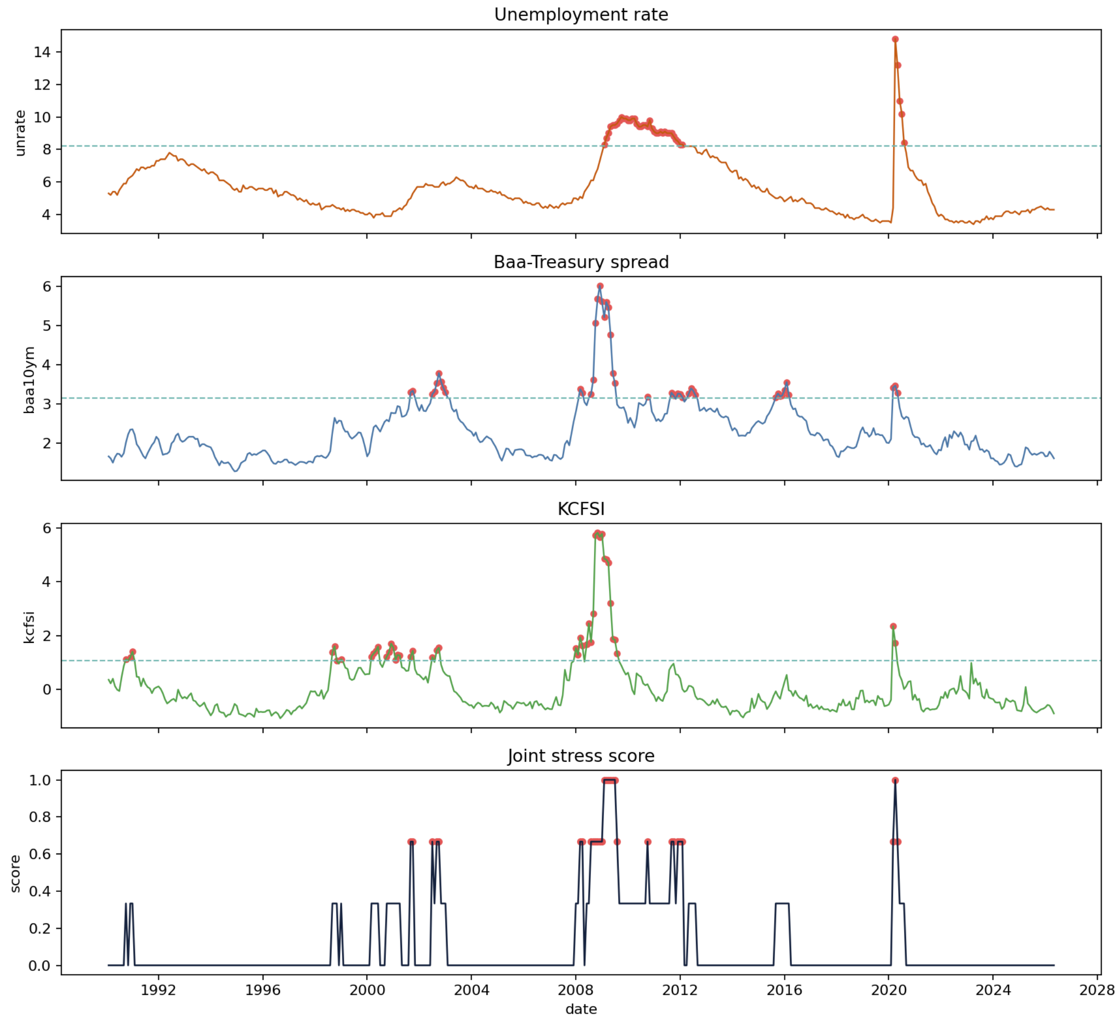

Figure 10: Empirical macro-financial illustration from the monthly FRED panel. The unemployment level exhibits highly concentrated upper-tail stress in a descriptive sense, while the credit spread and the financial-stress index recur across more separate episodes; the joint stress score peaks during the 2008–2009 crisis cluster.

7 Limitations and Scope

The limits of the paper are easiest to understand once its contribution is stated without inflation. The theoretical results are intentionally narrow. They establish interpretable properties of the cluster-adjusted stress functional and show how its behavior depends on the clustering term, but they do not amount to a new asymptotic theory for extremal-index estimation, a solution to threshold selection, or a general finite-sample theory for recurrence diagnostics. The paper therefore contributes an organizing functional and a set of elementary results around it, not a replacement for the deeper probabilistic literature on dependent extremes.

The empirical limits are equally important. The evidence presented here is primarily synthetic, which is appropriate for showing what the diagnostics do and do not measure, but insufficient for strong claims about macro-financial systems in the wild. Even in the small empirical illustration, the reported rankings are conditional on declared threshold, run-length, and sample-design choices rather than on a tuning-free structural notion of stress severity. A more mature empirical paper would therefore need defended threshold rules, uncertainty quantification, robustness analysis across run lengths and sample windows, and an interpretation anchored in a specific domain rather than in generic language about stress. Without those additions, the present manuscript should be read as methodological preparation for empirical work rather than as empirical resolution.

The multivariate component is narrower still. The stress-state index records when threshold-based stress is isolated and when it is joint, but it does not identify contagion, directional spillovers, or structural transmission channels. That limitation is not incidental. It marks the boundary between descriptive recurrence diagnostics and the larger enterprise of systemic-risk modeling.

These constraints should not be treated as apologetic caveats appended after the fact. They are part of the paper's intellectual discipline. The contribution is a methods and theory bridge for recurrence diagnostics in stress analysis, and it is more useful to state that contribution clearly, together with the thresholding and declustering choices on which it rests, than to overstate what the present evidence can bear.

8 Conclusion

The central argument of the paper is that stress analysis must distinguish tail magnitude from tail organization. Econometric and risk-measure literatures have developed powerful tools for the first object, but the second requires a more explicit treatment of recurrence, clustering, and return-time behavior than stress analysis usually receives. The framework developed here is meant to provide that treatment in a form that is mathematically legible and computationally reproducible.

Its contribution is best described plainly. The paper does not propose a new general theory of dependent extremes, nor does it resolve the empirical difficulties that arise when recurrence diagnostics are taken to real macro-financial data. What it does show is that recurrence can be formalized as a stress object, estimated through transparent pipeline components, summarized by an interpretable finite-threshold functional, and read alongside baseline volatility diagnostics without being absorbed by them. The synthetic evidence, together with the small empirical illustration, supports that claim under explicit thresholding and declustering designs rather than as a tuning-free structural ranking. The paper's value lies in stating the claim precisely enough that later empirical or theoretical work can either build on it or reject it on clear grounds.

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